

Cisco Secure Workload User Education Guide

Introduction, Video Library, and Onboarding Runbook

Cisco Secure Workload User Education

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Purpose

This guide helps users understand **Cisco Secure Workload (CSW)**, use a curated video library for learning, and follow a practical onboarding runbook for discovery, deployment, application dependency mapping, policy modeling, and staged micro-segmentation enforcement.

Disclaimer: This guide is **not** official Cisco product documentation. It is companion learning material for customer and partner education. Validate all design, scope, and feature support against your tenant's in-product documentation and Cisco product documentation before production use.

Use this as a user-friendly starting point. It is intentionally written in practical language, not as a replacement for Cisco product documentation, release notes, or design validation.

1. What Cisco Secure Workload Really Is

Cisco Secure Workload is a workload visibility and micro-segmentation platform. It helps customers understand how applications communicate, convert that understanding into policy, and enforce least-privilege communication across servers, virtual machines, cloud workloads, and supported application environments.

In simple terms, CSW answers four practical questions:

1. **What do I have?**

Inventory workloads, labels, operating systems, processes, and available metadata from agents, cloud connectors, CMDB systems, and other integrations.

2. **What talks to what?**

Observe L3 / L4 flows and, where supported, process-level context so teams can understand real application dependencies instead of guessing.

3. **What should be allowed?**

Use observed flows, labels, application groups, and policy modeling to recommend or build least-privilege segmentation policies.

4. **How do I enforce safely?**

Move from visibility to monitor mode, then to staged enforcement, with rollback and operational validation.

CSW is not just a firewall rule tool. It is an application dependency mapping, workload labeling, policy modeling, and micro-segmentation platform. The value comes from combining telemetry, labels, policy simulation, and controlled enforcement.

1.1 Micro-segmentation is a journey

Micro-segmentation is **not** something you create once and forget. Applications change — new microservices, refactors, cloud moves, patches, and integrations all shift real dependencies. Policy must stay aligned with that change or you either block legitimate traffic or slowly reopen paths you thought were closed.

CSW supports an ongoing program, not a one-off cutover:

- **Keep seeing the truth** — continuous flow and process telemetry (where enabled) so dependency maps reflect production, not a point-in-time snapshot.
- **Catch drift early** — monitor and simulation modes highlight new conversations before enforcement surprises an app owner.
- **Adapt policy deliberately** — ADM, labels, scopes, and modeling workflows exist to update allow rules when the business legitimately changes; vulnerability and forensics layers tighten or investigate when risk changes.
- **Operate for the long run** — change control, app-owner signoff, label sync from CMDB/cloud tags, API/CI/CD automation, and drift checks (see § 6.7 and § 8 Phase 8) are how teams sustain segmentation after the POV ends.

The phased roadmap in § 5 is a **on-ramp**. Phases 4–5 and day-2 operational practices are where journey-minded teams live after initial enforcement.

2. What CSW Solves: Lateral Movement and Ransomware

Cisco Secure Workload exists to break the assumption that an attacker who gets inside the network can move freely. Most enterprise networks are still relatively flat at the workload layer — once one server is compromised through a phishing link, an exposed admin port, a vulnerable web app, or a stolen credential, the attacker can usually reach hundreds of other servers, file shares, and databases over the same internal network paths that legitimate applications use.

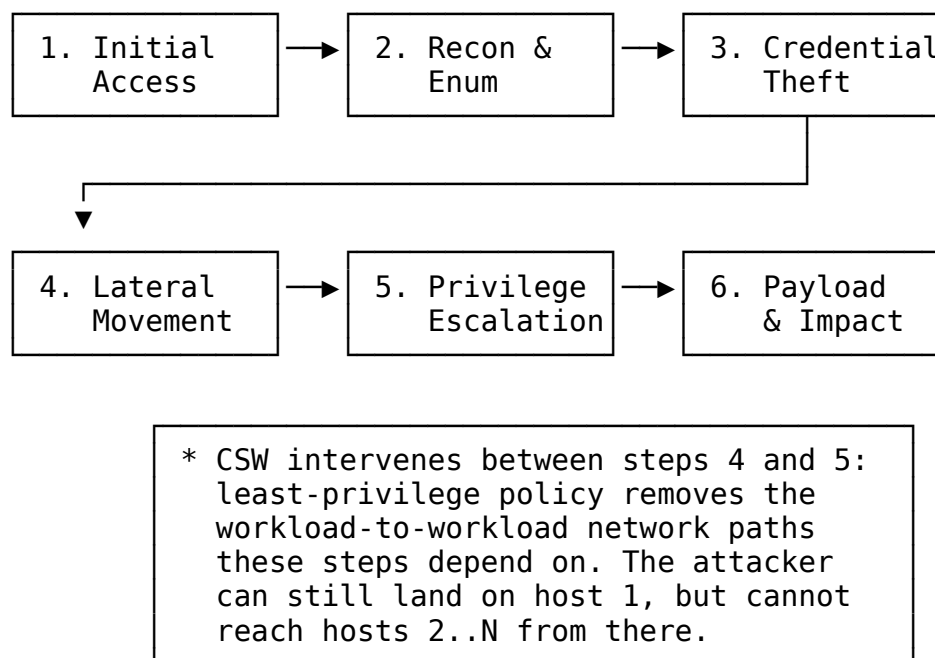
CSW reduces the **blast radius** of an intrusion by limiting each workload to **only the communication it actually needs**, based on real observed application behavior.

2.1 The Lateral Movement Problem

After initial access, modern attackers and most ransomware operators follow a predictable pattern:

1. **Land on one workload** — phishing, exposed RDP / SSH, exploited service, supply chain.
2. **Discover the internal network** — port scans, SMB enumeration, Active Directory reconnaissance.
3. **Steal credentials** — Mimikatz, LSASS dumps, Kerberoasting.
4. **Move laterally** — RDP, SMB, WinRM, WMI, PsExec, SSH, RPC, and other lateral admin tools.
5. **Escalate to high-value targets** — domain controllers, backup servers, file servers, databases, hypervisors.
6. **Stage and detonate the payload** — ransomware, data exfiltration, destruction.

Visually, the kill chain looks like this:



Steps 2 through 5 **all depend on the network allowing workload-to-workload traffic that no business application actually requires**. That is exactly the layer CSW controls.

2.2 How CSW Stops or Slows Ransomware

Ransomware groups make money by encrypting many systems in a short time window. They depend on:

- Unrestricted **SMB / file share access** between workstations and servers.
- Open **RDP / WinRM / WMI / PsExec** paths between servers.

- Reachable **backup servers, file servers, and hypervisors** from compromised endpoints.
- Free **east-west traffic** that lets the malware fan out before defenders can react.

CSW directly attacks each of those preconditions:

Ransomware behavior	What CSW does about it
SMB / RDP / WinRM fan-out from a compromised workload	Allow only application-required ports between specific workload groups; deny lateral admin protocols by default.
Mass file-share encryption	Restrict file-server access to the specific workloads, users, and processes that genuinely need it.
Reaching backup or recovery systems	Place backup servers in their own scope; permit traffic only from backup agents, not from general workloads.
Reaching domain controllers or identity tier	Scope the identity / Active Directory tier so only required clients and admin jump hosts can reach it.
Hopping between application tiers	Enforce per-app and per-tier policy (web → app → database) so a compromised web tier cannot reach unrelated databases.
Spreading from dev / test / lab into prod	Hard-segment prod from non-prod so a non-prod compromise cannot pivot into prod.
Long undetected dwell time	Surface every blocked or anomalous flow as evidence for SOC and incident response.

The result is that **ransomware that lands on one workload finds the network around it almost empty**: the protocols it needs to spread are blocked by policy, and every attempt is logged.

2.3 Why We Need It

- **Flat networks no longer match the threat model.** Perimeter firewalls do not stop an attacker who is already inside.
- **Identity-based and EDR controls are necessary but not sufficient.** They catch behavior on the host. CSW removes the network paths the attacker would use between hosts.
- **Crown-jewel applications need explicit protection.** Payments, claims, customer data, intellectual property, and backup infrastructure should not be reachable from a random user workstation or a low-tier dev server.
- **Compliance and audit demand it.** PCI, HIPAA, SOX, and most internal security frameworks expect documented segmentation between regulated and non-regulated

systems. For framework-by-framework mappings — customer-facing reports and matching SA / SE technical runbooks across HIPAA, SOC 2, PCI DSS v4, NIST 800-53, ISO 27001:2022, CISA ZTMM, FIPS 140, NIST 800-207 / 207A, DORA, NIS2, NERC CIP, TSA Pipeline, CIS Controls v8.1, NIST CSF 2.0, CMMC 2.0, and more — see the companion repository: [chandrapati/CSW-Compliance-Mapping](#). Use it whenever a customer asks "how does CSW map to <framework>?".

- **It must not break applications.** CSW's discovery-first model (map dependencies → label workloads → model policy → enforce in stages) is what makes segmentation finally feasible in real enterprises.

In one sentence: **CSW exists so that when — not if — one workload is compromised, the attacker cannot reach the next one.**

3. Why Customers Care

Customers usually evaluate CSW because they need to reduce lateral movement risk without breaking applications. Traditional network segmentation often depends on VLANs, subnets, firewall zones, or static IP lists. That model becomes difficult when applications span data centers, public cloud, containers, and shared services.

CSW helps customers:

- Discover application dependencies before enforcing policy.
 - Build policies based on labels and application intent, not only IP addresses.
 - Protect high-value or vulnerable applications with tighter controls.
 - Align policy with CMDB data, cloud tags, application ownership, and observed behavior.
 - Roll out segmentation in phases, starting with visibility and policy modeling.
 - Produce evidence for security, audit, and architecture teams.
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4. Core CSW Concepts for Users

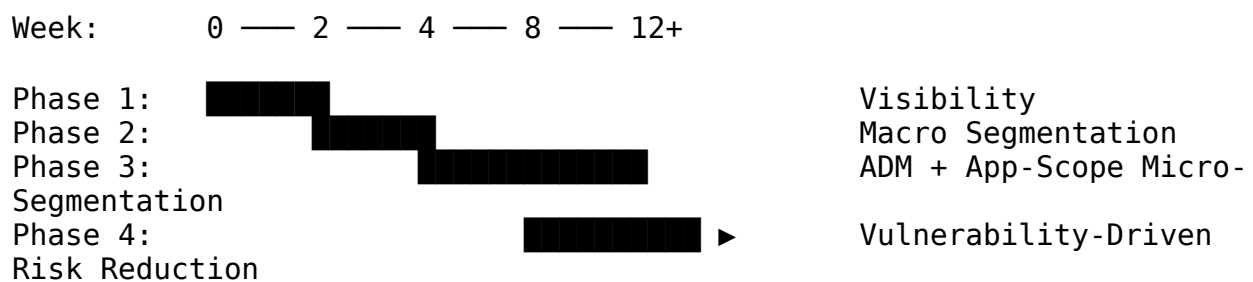
Concept	User-friendly explanation
Workload	A server, VM, cloud instance, container host, or supported compute asset being observed or protected.
Agent / enforcer	Software installed on supported workloads to collect telemetry and, when enabled, enforce policy.
Flow visibility	The observed network communication between workloads, usually source, destination, protocol, port, and time.
Process visibility	Mapping a network flow to the local process or service using it, where

Concept	User-friendly explanation
	supported.
Labels	Metadata used to describe workloads, such as application, environment, role, owner, location, or risk.
Scope	A logical boundary for organizing workloads and policies.
Policy modeling	Previewing what would be allowed or blocked before enforcing.
Monitor mode	Observe and report policy behavior without blocking traffic.
Enforcement mode	Actively allow or deny traffic based on policy.
Connectors	Integrations that bring in context from systems such as ServiceNow, cloud providers, or other sources.
Edge appliance	A CSW virtual appliance used to run certain connectors, such as the ServiceNow connector.
Ingest	A pipeline or appliance pattern used where external telemetry ingestion is required. Validate design with Cisco for the customer use case.

5. Phased Adoption Roadmap

CSW value compounds in phases. You do **not** need to wait for full Application Dependency Mapping (ADM) to get value — each phase below delivers a concrete outcome on its own, and a customer who stops after Phase 2 has already shrunk ransomware blast radius and isolated prod from non-prod.

This section describes the **value-tier view** of adoption: what outcomes show up when. The tactical, step-by-step deployment playbook (tenant prep, agent rollout, label strategy, policy modeling, enforcement testing, operationalization) lives in **§ 8 CSW Onboarding Runbook**.





Phase	Window	What you do	What you ship	Primary value
1 — Visibility	Week 0–2	Deploy agents on a representative slice. Turn on cloud, CMDB, identity, and DNS connectors. Import existing labels.	Workload inventory, label dictionary, L3 / L4 + process flow data, baseline dependency view.	See what you actually have, with real flow truth — often the first time application owners can answer "who does this server talk to?".
2 — Macro Segmentation	Week 2–4	Apply broad zone-to-zone policy in monitor mode, then enforcement: prod ↔ non-prod, prod ↔ dev / test / staging, user VLANs ↔ server zones. Deny lateral admin protocols (SMB, RDP, WinRM, WMI, PsExec, SSH) where no business app needs them. Isolate the backup tier and identity tier.	Enforced macro policy with monitor-mode evidence and rollback plan.	Biggest blast-radius reduction for the lowest effort. Ransomware fan-out paths are cut. The prod / non-prod story holds up in audit. High-value tiers (backup, AD, hypervisor) are reachable only from where they should be.
3 — ADM + App-Scope Micro-	Week 4–12 (per application	Run ADM on a chosen application.	Per-app dependency map, modeled	True micro-segmentation around the

Phase	Window	What you do	What you ship	Primary value
Segmentation	wave)	Review the dependency map with the app owner. Convert observed flows + labels into recommended allow rules. Move to monitor mode, tune, then staged enforcement. Repeat for the next app wave.	policy, blocked-flow evidence, allowed-business-transaction evidence, app-owner signoff.	application itself. Per-tier policy (web → app → db). App teams understand their own application, sometimes for the first time.
4 — Vulnerability-Driven Risk Reduction	Continuous from ~Week 8	Ingest vulnerability data (scanner exports, CVE feeds). Tag workloads with risk labels (for example <code>risk:high</code> , <code>cve:exploitable</code>). Tighten policy automatically for high-risk workloads — restrict their reachable surface to admin / patch paths only until remediated.	Risk-tagged inventory, tightened policy on vulnerable workloads, security and audit evidence.	Defenders work the problems attackers actually try. When the next Log4J / Log4Shell-class CVE lands, exposure can be shrunk within the same day instead of waiting on patch cycles.
5 — Forensics	Continuous from ~Week	Use Secure Workload	Forensics events,	Detection compounds

Phase	Window	What you do	What you ship	Primary value
and Anomaly Detection	8	forensics events, flow-pattern anomaly detection, and SIEM / SOAR integration. Build playbooks that pair policy violations with host evidence.	anomaly findings, SOC playbooks, IR evidence trails.	with segmentation. Every blocked flow becomes evidence. Mean-time-to-detect drops because workload behavior is bounded by policy.

Key idea: every phase is independently valuable. A customer who stops after Phase 2 still wins — ransomware fan-out is gone and prod is isolated from non-prod. A customer who reaches Phase 5 has continuous defense in depth — and should keep operating there, because applications and policy both keep changing (see § 1.1).

Mapping phases to videos in the catalog

- **Phase 1 — Visibility:** Agent Configuration Profile (#1), Scopes (#3), Labels (#4), Inventory Filters (#5), Flow Analysis (#10).
- **Phase 2 — Macro Segmentation:** SSH Risk Reduction (#15), Terminal Services Segmentation (#16), Production and Test Risk Reduction (#6), VDI Segmentation (#17).
- **Phase 3 — ADM + App-Scope Micro:** Application Dependency Mapping & Policy Analysis (#7), Policy Visual and Quick Analysis (#8), Dynamic Workloads & Policy (#9), AI-Driven Policy Suggestions (#33), Policy Statistics with AI Engine (#34).
- **Phase 4 — Vulnerability-Driven Risk Reduction:** Vulnerabilities and Risk Reduction (#11), Log4J Risk Reduction (#14), Security Dashboard (#12).
- **Phase 5 — Forensics and Anomaly Detection:** Forensics (#13), Flow Analysis (#10), Security Dashboard (#12).

6. CSW Video Library (Learning Path Order)

Videos are listed in the order that builds CSW skills fastest — not grouped by product theme. **Watch top to bottom** within each module. The **Catalog #** column matches the original Cisco slide numbering (used in § 5 phase mappings).

6.1 Module 1 — Foundations

Watch	Catalog #	Video	Description	Link
1	3	Scopes	Group workloads logically for policy.	Watch
2	4	Labels	Tag workloads for policy and visibility.	Watch
3	5	Inventory Filters	Focus inventory on specific workload groups.	Watch

6.2 Module 2 — Agent deployment

Watch	Catalog #	Video	Description	Link
4	1	Agent Configuration Profile	Configure agent profiles for enforcement.	Watch
5	2	Golden Image VDI	VDI golden-image agent deployment. <i>Optional.</i>	Watch

6.3 Module 3 — Visibility and dependency discovery

Watch	Catalog #	Video	Description	Link
6	6	Production and Test Risk Reduction	Macro-segment prod vs non-prod.	Watch
7	10	Flow Analysis	Traffic patterns and anomalies.	Watch
8	7	Application Dependency Mapping & Policy Analysis	ADM and policy derivation.	Watch
9	9	Dynamic Workloads & Policy	Policy for moving/scaling workloads.	Watch

Watch	Catalog #	Video	Description	Link
10	8	Policy Visual and Quick Analysis	Visualize policy impact.	Watch

6.4 Module 4 — AI-assisted policy

Watch	Catalog #	Video	Description	Link
11	33	AI-Driven Policy Suggestions	AI policy recommendations.	Watch
12	34	Policy Statistics with AI Engine	Statistics and tuning at scale.	Watch

6.5 Module 5 — Security, risk, and forensics

Watch	Catalog #	Video	Description	Link
13	12	Security Dashboard	Posture and risk drill-downs.	Watch
14	11	Vulnerabilities and Risk Reduction	Vulnerable workload prioritization.	Watch
15	15	SSH Risk Reduction	Control SSH between workloads.	Watch
16	14	Log4J Risk Reduction	Log4Shell exposure containment.	Watch
17	13	Forensics	Investigation evidence.	Watch

6.6 Module 6 — Segmentation use cases

Watch	Catalog #	Video	Description	Link
18	16	Terminal Services Segmentation	RDS / Terminal Services.	Watch
19	17	VDI Segmentation	Golden-image VDI estates.	Watch

6.7 Module 7 — Integrations (POV-specific)

Watch	Catalog #	Resource	Description	Link
20	18	Secure Workload & Firewall Integration (Part 1)	Introduction, design, architecture.	Watch
21	19	Secure Workload & Firewall Integration (Part 2)	Deployment patterns and policy flow.	Watch
22	20	Secure Workload & Firewall Integration (Part 3)	Enforcement, telemetry, operations.	Watch
23	21	Secure Firewall White Paper	Joint architecture (Cisco.com).	Read
24	22	Firewall Integration Deep Dive	Design principles and use cases.	Read
25	23	F5 BIG-IP: APM Visibility	F5 APM application visibility.	Watch
26	24	F5 BIG-IP AFM	AFM flow context.	Watch
27	25	F5 BIG-IP IPFIX Configuration	IPFIX into Secure Workload.	Watch
28	26	DNS Server Integration	DNS context for flows and policy.	Watch
29	27	Infoblox Integration	DNS / IPAM context.	Watch
30	28	Algosec Integration	Firewall-policy lifecycle.	Watch
31	29	ISE (In Action)	Identity from Cisco ISE.	Watch
32	30	FMC / Edge /	FMC	Watch

Watch	Catalog #	Resource	Description	Link
		Ingest / Appliance	integration paths.	
33	31	ACI and CSW Integration	ACI + workload segmentation.	Watch
34	35	Splunk Integration (SIEM)	Cisco Security Cloud App for baseline datasets, CSW → Splunk Syslog alerts, plus Splunk-driven Python against the CSW API.	Watch
35	36	CI/CD Pipeline Integration	Labels, scopes, and policies as code in git; pipeline-driven API calls keep the CSW tenant in sync.	Watch

Splunk integration — three patterns to plan for in a POV:

#	Pattern	What it gives the customer	Effort
1	Cisco Security Cloud App for Splunk	Install the Cisco-published Splunk app; it reaches back to CSW for a baseline set of datasets and dashboards. Demonstrates value with minimal configuration.	Lowest
2	CSW → Splunk Syslog alerts	Configure CSW to push alarms /	Low

#	Pattern	What it gives the customer	Effort
		alerts to Splunk over Syslog. SOC sees policy violations, agent-health events, and forensic signals next to the rest of their telemetry.	
3	Splunk → CSW API (Python)	Splunk launches Python scripts against the CSW API. Anything the API exposes — inventory, scopes, labels, policies, enforcement state, vulnerabilities, flow summaries (everything except the raw flow data itself) — can be pulled into Splunk and indexed, dashboarded, or alerted on. Any CSW API automation already written becomes a Splunk-driven data feed.	Medium → High (custom)

POV positioning: start with pattern 1 to prove the integration path is live, layer in pattern 2 so the SOC has alerts in their existing console, and reserve pattern 3 for the "custom dataset" conversation where the customer asks for something the out-of-the-box app does not surface. The pattern 3 ceiling is essentially the CSW API surface itself.

CI/CD pipeline integration — what it typically wires up:

Wire-up	What it gives the customer	Where it lands in the runbook
Policy-as-code —	Tenant changes go	§ 8 Phase 8

Wire-up	What it gives the customer	Where it lands in the runbook
scopes, labels, intent-policy stored in git; pipeline applies via CSW API.	through code review, not console clicks. Auditable, reviewable, revertable.	(Operationalize) — replaces ad-hoc UI changes.
Promotion across environments — same bundle rolls dev → staging → prod, starting in Monitor mode.	Lower-risk roll-forward; the production change is the same artifact that already passed in lower environments.	§ 8 Phase 6 (Monitor Mode) → Phase 7 (Enforce).
Label sync from sources of truth — CMDB / cloud tag exports normalized in CI and pushed to CSW.	Labels stay accurate over time without manual upkeep, which is the #1 way label strategies decay.	§ 8 Phase 2 (Build the Label Strategy).
Drift detection — scheduled job diffs live tenant vs git.	Out-of-band changes get caught and either backported to git or rolled back.	§ 8 Phase 8 (Operationalize) — change-process enforcement.
Same API surface as Splunk pattern 3 — Python you already wrote for CSW becomes a pipeline step.	Investment in CSW API automation compounds across SIEM ingestion and CI/CD.	§ 8 Phase 8 — "Build API or CI/CD automation where needed."

POV positioning: the CI/CD story usually lands well with cloud / platform teams who already manage everything else as code and bristle at "click here in the console" instructions. Start with policy-as-code for one application scope from the POV — it pairs naturally with the Phase 3 ADM → policy modeling output, since the modeled policy is already structured data.

6.8 Module 8 — Containers and Kubernetes

Watch	Catalog #	Video	Description	Link
36	32	Agent K8s	Agent in Kubernetes.	Watch

7. How to Explain CSW in a Customer Meeting

30-second version

Cisco Secure Workload helps customers discover how applications communicate and then safely enforce least-privilege segmentation policies around those applications. It starts with visibility and dependency mapping, uses labels and observed behavior to create policy, and supports staged enforcement so teams can reduce lateral movement risk without breaking applications.

2-minute version

Most enterprises want micro-segmentation, but they hesitate because they do not fully know what their applications depend on. CSW starts by collecting workload telemetry, flow visibility, process context, cloud metadata, and CMDB labels. That data helps application teams understand real dependencies. From there, teams can model policy, see what would be blocked, tune the rules, and move selected workloads from monitor mode to enforcement. CSW is especially useful when customers need consistent segmentation across data center and cloud environments.

Common positioning

- **For security:** Reduce lateral movement and contain high-risk applications.
 - **For network teams:** Shift from static IP / subnet rules to workload-aware policy.
 - **For application teams:** Discover dependencies before enforcement and avoid unexpected outages.
 - **For cloud teams:** Use cloud tags and dynamic metadata so policy follows workloads.
 - **For operations:** Start in monitor mode, model changes, and roll out enforcement gradually.
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8. CSW Onboarding Runbook

Phase 0 - Confirm Scope and Outcomes

Goal: Agree on what the customer wants to prove.

Steps:

1. Identify the target customer environment: data center, public cloud, containers, or a mix.
2. Select 2-3 representative applications.
3. Pick at least one low-risk application and one higher-value application.
4. Define success criteria:
 - Flow visibility demonstrated.
 - Process visibility demonstrated.
 - Application dependency map reviewed by app owner.
 - Labels imported or created.
 - Policy modeled before enforcement.
 - One controlled enforcement test completed.

Output: POV scope, application list, stakeholder list, and success criteria.

Phase 1 - Prepare the Tenant and Access

Goal: Prepare the CSW management environment and user access.

Steps:

1. Confirm CSW tenant or cluster access.
2. Create required administrator, operator, and viewer accounts.
3. Configure SSO / MFA / RBAC if in scope.
4. Confirm API access if automation or reporting is part of the POV.
5. Document who can create labels, scopes, policies, and enforcement changes.

Output: Access matrix, RBAC model, and tenant readiness confirmation.

Phase 2 - Build the Label Strategy

Goal: Establish labels before policy design.

Recommended starting labels:

Label type	Example values	Source
Application	payments, claims, customer-portal	CMDB, app owner, manual
Environment	prod, non-prod, dev, test	CMDB, cloud tag
Role	web, app, db, middleware	CMDB, observed behavior
Location	dc1, dc2, aws, azure	Cloud connector, manual
Owner	team-a, team-b	CMDB
Risk	high, medium, low	vulnerability scanner, security team

Steps:

1. Agree on a minimal label taxonomy.
2. Map sources for each label: ServiceNow, cloud tags, manual, API, or observed behavior.
3. Avoid creating too many labels before discovery. Start simple.
4. Validate that labels can support the first policy examples.

Output: Initial label dictionary and source-of-truth mapping.

Phase 3 - Deploy Agents and Connectors

Goal: Start collecting telemetry and context.

Steps:

1. Install CSW agents on selected Linux / Windows workloads.

2. Confirm agent health and visibility.
3. Deploy a CSW Edge virtual appliance if connector workloads are required.
4. Configure ServiceNow connector if CMDB labels are in scope.
5. Configure cloud connectors if public cloud inventory / tags are in scope.
6. Configure Ingest only if external flow or metadata ingestion is required and validated by Cisco design.

Output: Healthy agents, connector status, imported labels, and visible workload inventory.

Phase 4 - Discover Application Dependencies

Goal: Learn how the application actually communicates.

Steps:

1. Let workloads run through normal business transactions.
2. Capture L3 / L4 flow data.
3. Review process-level visibility where supported.
4. Build application dependency maps.
5. Review dependencies with application owners.
6. Identify unexpected flows, shared services, admin paths, and risky protocols.

Output: Dependency map, flow summary, process evidence, and app-owner review notes.

Phase 5 - Model Policy

Goal: Build policy without breaking the application.

Steps:

1. Use observed flows and labels to create recommended allow rules.
2. Group policy by application tiers where possible.
3. Identify required shared services such as DNS, NTP, authentication, logging, monitoring, backup, and patching.
4. Model the policy before enforcement.
5. Review predicted blocked flows with application and operations teams.
6. Tune policy until the expected business transactions are allowed.

Output: Proposed policy, modeled impact, predicted denies, and approval to test enforcement.

Phase 6 - Start in Monitor Mode

Goal: Validate policy behavior before blocking.

Steps:

1. Apply policy in monitor or non-blocking mode where supported.
2. Run application test transactions.
3. Review policy violations and unexpected flows.
4. Update labels or policy rules as needed.
5. Confirm application owners are comfortable with the modeled impact.

Output: Monitor-mode results, policy tuning notes, and enforcement readiness decision.

Phase 7 - Enforce in a Controlled Scope

Goal: Prove micro-segmentation safely.

Steps:

1. Select a small enforcement scope.
2. Schedule a change window.
3. Confirm rollback plan.
4. Move selected workloads to enforcement.
5. Test allowed traffic.
6. Test intentionally denied traffic.
7. Confirm logs, alarms, and reports show the expected result.
8. Monitor application health after enforcement.

Output: Enforcement proof, blocked-flow evidence, allowed-flow evidence, rollback plan, and operational signoff.

Phase 8 - Operationalize

Goal: Make CSW sustainable after the POV.

Steps:

1. Define ownership for labels, scopes, and policies.
2. Document policy change process.
3. Integrate logs / alarms with SIEM or SOAR if in scope.
4. Create recurring policy review cadence.
5. Build API or CI/CD automation where needed. *(See § 6.7 Module 7 — Integrations for the CI/CD pipeline patterns: policy-as-code, environment promotion, label sync, and drift detection.)*
6. Define exception handling for unsupported workloads.
7. Expand from the initial application set to the next wave.

Output: Operating model, change process, exception process, and expansion roadmap.

9. Discovery Questions

Use these questions before sizing or designing a POV:

1. Which business application should be used for the first dependency map?
 2. Which application owner can validate the map?
 3. Which environments are in scope: prod, non-prod, dev, test?
 4. Which platforms are in scope: VMware, bare metal, public cloud, containers?
 5. Which operating systems are in scope?
 6. Is ServiceNow the CMDB source of truth?
 7. Are cloud tags clean enough to use as labels?
 8. Is MFA / SSO required for CSW administrator access?
 9. Are agents allowed on the selected workloads?
 10. Is enforcement part of the POV, or visibility-only?
 11. What is the rollback process for policy enforcement?
 12. Which SIEM / SOAR tools should receive alarms or logs? (*If Splunk: confirm interest in the Cisco Security Cloud App, Syslog alert forwarding, and/or Python-driven API ingestion — see § 6.7 Module 7.*)
 13. What vulnerability scanner or risk source should be used for high-risk labels?
 14. Who approves policy changes?
 15. What evidence does the customer need at the end of the POV?
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10. POV Evidence Checklist

Evidence	Why it matters
Agent health screenshot	Proves telemetry source is active
Workload inventory screenshot	Proves assets are visible
Label inventory screenshot	Proves metadata is usable
ServiceNow connector status	Proves CMDB integration path
Cloud connector status	Proves cloud context ingestion
Flow table export	Proves L3 / L4 visibility
Process-level flow screenshot	Proves process context where supported
Application dependency map	Proves discovery value
Policy recommendation	Proves policy can be derived from observed behavior
Policy modeling output	Proves change impact can be reviewed before enforcement
Monitor-mode policy result	Proves safe pre-enforcement workflow

Evidence	Why it matters
Enforcement blocked-flow test	Proves actual segmentation control
Allowed business transaction test	Proves enforcement did not break required traffic
Audit log	Proves administrative traceability
API output	Proves automation path
SIEM event	Proves operational integration (for Splunk: Cisco Security Cloud App dashboard panel, a CSW Syslog alert in the indexer, or a Python-pulled CSW API dataset)

11. Common Pitfalls

Pitfall	How to avoid it
Starting with enforcement too early	Start with discovery, labels, and modeling before blocking traffic.
Using too many labels too soon	Start with application, environment, role, location, owner, and risk. Expand later.
Skipping application owner validation	Dependency maps must be reviewed by the app owner before policy.
Ignoring shared services	Always account for DNS, NTP, identity, logging, monitoring, backup, and patching.
Treating cloud and data center separately	Use consistent labels across both so policy is portable.
Forgetting rollback	Every enforcement test needs a rollback plan and owner.
Overclaiming integration support	Confirm exact platform and version support with current Cisco documentation.

12. Simple CSW Talk Track

Use this flow in customer conversations:

1. **Start with the risk:** Attackers move laterally after initial compromise. Flat networks make that easier.
2. **Explain the blocker:** Customers hesitate to segment because they do not know every application dependency.

3. **Position CSW:** CSW discovers dependencies first, then models policy, then enforces gradually.
 4. **Make it practical:** We start with a few representative apps, import labels, review flows, model policy, and only then enforce.
 5. **Close with outcome:** The customer gets application maps, policy recommendations, enforcement proof, and an expansion roadmap.
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13. Companion Repositories for Deeper Learning

This guide is the educational front door. Once a reader is comfortable with the concepts and the video library, the repositories below pick up where this guide ends. They are listed in the order most learners hit them: install agents, run the policy lifecycle, then map it all to compliance frameworks.

Repo	When you reach for it
chandrapati/CSW-Agent-Installation-Guide	Right after this guide, when it is time to actually install and operate the CSW host agent on Linux, Windows, cloud, container, or agentless workloads.
chandrapati/CSW-Policy-Lifecycle	When you move from "I understand policy" (covered here) to running the full lifecycle in a tenant: discovery (ADM) → analysis → enforcement (Monitor → Simulate → Enforce) → day-2 operations.
chandrapati/CSW-Compliance-Mapping	When a customer or auditor asks "how does CSW support framework X?" — covers HIPAA, SOC 2, PCI DSS v4, NIST 800-53, ISO 27001:2022, CISA ZTMM, FIPS 140, NIST 800-207/207A, DORA, NIS2, NERC CIP, TSA Pipeline, CIS Controls v8.1, NIST CSF 2.0, CMMC 2.0, and more.

For the full list of related Cisco Secure Workload repositories — including POV templates, blast-radius demo, tenant insights, and SE helpers — see the **Related Cisco Secure Workload Repositories** section in this repo's [README](#).
